Two fits. One limit.

Saved application-of-areas experiment: falling short, given line 12, target area 28, trial 792893/250000.

The coordinates, trial numbers, graph and square-completion identities are modern teaching work. The readings are selected passages, not complete transcriptions of the three proofs. The whole printed pages retain the complete arguments and editorial apparatus. Heiberg’s Latin is a nineteenth-century translation; the English and language guidance are new and have not had independent philological review.

Your trial and the prescribed shape

Your trial · t

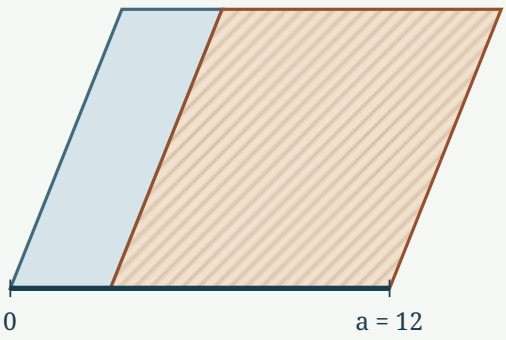
t ≈ 3.171572
Applied area ≈ 28.00000



height ≈ 3.171572
excluded shortfall area ≈ 10.05887

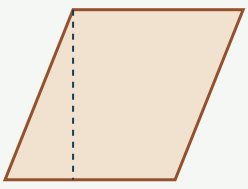
Reflected trial · a − t

t ≈ 8.828428
Applied area ≈ 28.00000



height ≈ 8.828428
excluded shortfall area ≈ 77.94114

Given shape · height 1



Blue is the part within the line; hatching is the excluded shortfall. The plates use equal horizontal and vertical units. The shape’s k is vertical height divided by base. Shear changes the given shape; at each fixed shear, all hatched regions are scaled copies of that shape.

This trial does not fit exactly

Applied area **1749999690551/62500000000** < target **28**.

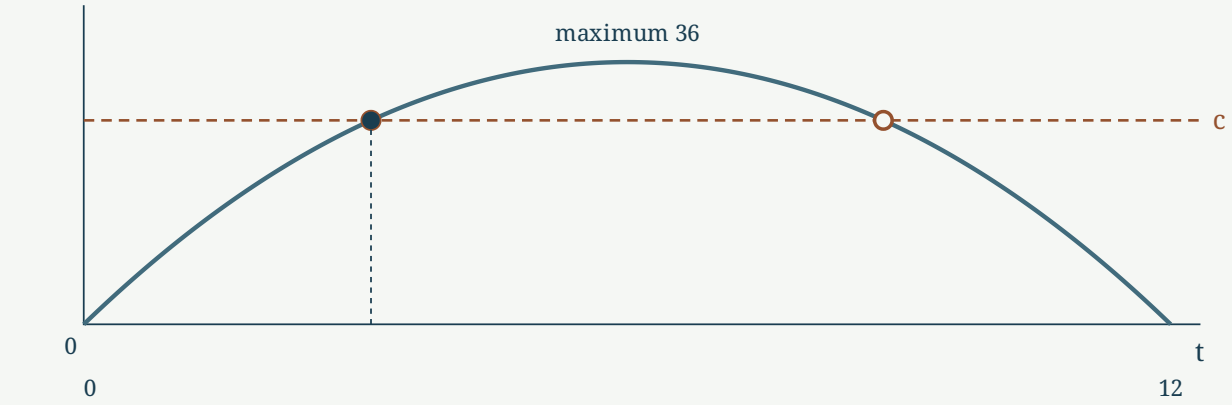
Area minus target = **-309449/62500000000**. No tolerance or rounded decimal turns a near miss into equality.

Exact area account · arbitrary length and area units

Quantity	Value	Meaning
a	12	Given line
t	792893/250000	Shortfall base
a − t	2207107/250000	Applied base
k·t	792893/250000	Perpendicular height, not slant side
Applied area	1749999690551/62500000000	k·t(a − t)
Similar region	628679309449/62500000000	Excluded shortfall; k·t²
c	28	Target area
Area − c	-309449/62500000000	Zero means exact fit

Possible fits

Area as t changes · modern graph



Two fits: target 28 is below the maximum 36.

1. **792893/250000 < t < 3171573/1000000**
   Irrational fit; approximately 3.17157288. The rational endpoints bracket the root on this branch.

2. **8828427/1000000 < t < 2207107/250000**
   Irrational fit; approximately 8.82842712. The rational endpoints bracket the root on this branch.

For these coordinates, t = (a ± √D)/2, with D = a² − 4c/k = 32. Only 0 < t < a counts; when D = 0 the two expressions coincide. Square roots are modern notation. Certified fractions determine every displayed interval.

Why there is a ceiling

$$k \cdot a^2/4 - k \cdot t(a - t) = k \cdot (t - a/2)^2$$

$$36 - 1749999690551/62500000000 = 500000309449/62500000000$$

The right side cannot be negative. It is zero only at t = a/2, so that cut gives the unique maximum. Replacing t by a − t keeps the product t(a − t) unchanged: the two trial panels have equal areas.

This algebra is a modern explanation of this family. Euclid uses similar figures, equal complements and gnomons. Read the selected source steps below; earlier constructions such as VI.25 remain prerequisites.

Read the construction conditions

Nine selected readings with eighteen contextual word notes. These do not replace the complete source pages below.

VI.28 · opening problem, first clause

1. Apply, while falling short

Greek · p.162

Παρὰ τὴν δοθεῖσαν εὐθεῖαν τῷ δοθέντι εὐθυγράμμῳ ἴσον παραλληλόγραμμον παραβαλεῖν ἐλλεῖπον εἴδει παραλληλογράμμῳ ὁμοίῳ τῷ δοθέντι·

Heiberg’s Latin · p.163

Datae rectae datae figurae rectilineae aequale parallelogrammum adplicare deficiens figura parallelogramma datae simili.

To apply to the given straight line a parallelogram equal to the given rectilinear figure, falling short by a parallelogram similar to the given one.

The line, target figure and shape of the shortfall are three different givens. ἴσον concerns area; ὁμοίῳ concerns shape. In the plate, a names the line, c the target area, and the hatched region stays similar to the displayed given shape. Our a, c and t are modern labels, not the source’s point names.

παραβαλεῖν / adplicare
to apply or lay alongside; an infinitive stating the construction problem

ἐλλεῖπον / deficiens
falling short; the participle describes the parallelogram to be made

VI.28 · opening problem, restriction

2. The condition belongs to the problem

Greek · p.162

δεῖ δὲ τὸ διδόμενον εὐθύγραμμον [ᾧ δεῖ ἴσον παραβαλεῖν] μὴ μεῖζον εἶναι τοῦ ἀπὸ τῆς ἡμισείας ἀναγραφομένου ὁμοίου τῷ ἐλλείμματι [τοῦ τε ἀπὸ τῆς ἡμισείας καὶ ᾧ δεῖ ὅμοιον ἐλλείπειν].

Heiberg’s Latin · p.163

oportet autem, figuram rectilineam datam maiorem non esse figura in dimidia recta descripta defectui simili.

But the given rectilinear figure [to which an equal figure must be applied] must not be greater than the figure drawn on half the line, similar to the shortfall [both the one on half the line and the one to which the shortfall must be similar].

μὴ μεῖζον / maiorem non means ‘not greater’, so equality is allowed. Square brackets are Heiberg’s editorial brackets, retained here; the final bracketed Greek explanation is not separately repeated in his Latin. The target 45 exceeds 36 for a = 12 and k = 1. That is a failed premise, not a failure to move the control precisely enough.

μὴ μεῖζον / maiorem non
not greater; includes equality

τῆς ἡμισείας / in dimidia
of or on the half; the understood object is the given straight line

VI.27 · conclusion

3. The greatest is on the half

Greek · p.160

Πάντων ἄρα τῶν παρὰ τὴν αὐτὴν εὐθεῖαν παραβαλλομένων παραλληλογράμμων καὶ ἐλλειπόντων εἴδεσι παραλληλογράμμοις ὁμοίοις τε καὶ ὁμοίως κειμένοις τῷ ἀπὸ τῆς ἡμισείας ἀναγραφομένῳ μέγιστόν ἐστι τὸ ἀπὸ τῆς ἡμισείας παραβληθέν· ὅπερ ἔδει δεῖξαι.

Heiberg’s Latin · p.161

Ergo omnium parallelogrammorum eidem rectae adplicatorum et deficientium figuris parallelogrammis similibus et similiter positis ei, quae in dimidia describitur, maximum est, quod dimidiae adplicatur; quod erat demonstrandum.

Therefore, of all the parallelograms applied to the same straight line and falling short by parallelograms similar and similarly placed to the one drawn on half the line, the greatest is the one applied on half the line; which was to be proved.

Euclid compares regions through equal complements and a gnomon. Our identity maximum − trial = k(t − a/2)² explains this coordinate family in modern algebra. It is not a transcription of Euclid’s proof. Heiberg’s p.161 note discusses an additional manuscript case that he removed to the appendix. The app’s reflected trial is a modern operation, not a claim that this edition presents two separate constructions.

μέγιστον / maximum
greatest, the superlative rather than merely ‘greater’

ὁμοίως κειμένοις / similiter positis
similarly placed; the shortfall cannot be an arbitrary shape of the right area

VI.28 · equal case after bisection

4. If it is equal, the job is done

Greek · p.164

Εἰ μὲν οὖν ἴσον ἐστὶ τὸ ΑΗ τῷ Γ, γεγονὸς ἂν εἴη τὸ ἐπιταχθέν·

Heiberg’s Latin · p.165

iam si AH = Γ, effectum erit propositum.

If, then, AH is equal to Γ, what was required would have been accomplished.

The source has already bisected AB and constructed a similar figure on the half. Here AH and Γ name areas, not lengths. The equality case is part of the construction: there is no positive difference left to construct. With the opening shape, change the target to 36 and set t = 6.

εἰ μὲν / si
if, introducing the first of the cases; εἰ δὲ οὔ follows in the source

τὸ ἐπιταχθέν / propositum
what was required or proposed; a substantivized participle

VI.28 · smaller target

5. Construct the difference with the same shape

Greek · p.164

ᾧ δὴ μεῖζόν ἐστι τὸ ΗΒ τοῦ Γ, ταύτῃ τῇ ὑπεροχῇ ἴσον, τῷ δὲ Δ ὅμοιον καὶ ὁμοίως κείμενον τὸ αὐτὸ συνεστάτω τὸ ΚΛΜΝ.

Heiberg's Latin · p.165

iam excessui, quo maius est HB figura Γ, aequale et parallelogrammo Δ simile et similiter positum idem construatur ΚΛΜΝ.

Construct ΚΛΜΝ equal to the excess by which HB is greater than Γ, and at the same time similar and similarly placed to Δ.

There are two constraints on one new figure: its area is a difference, and its shape is prescribed. VI.25 supplies that construction. For a square shortfall our modern normalized equation is $(t - a/2)^2 = a^2/4 - c/k$. If the right side is negative, there is no real length to construct.

ὑπεροχῇ / excessui
to the excess or difference; here the amount by which one area exceeds another

τὸ αὐτό / idem
the same figure must satisfy the area and similarity requirements together

VI.28 · first remainder step

6. Equal remainders

Greek · p.166

ἐπεὶ οὖν ἴσον ἐστὶ τὸ ΒΗ τοῖς Γ, ΚΜ, ὧν τὸ ΗΠ τῷ ΚΜ ἐστιν ἴσον, λοιπὸς ἄρα ὁ ΥΧΦ γνώμων λοιπῷ τῷ Γ ἴσος ἐστίν.

Heiberg’s Latin · p.167

iam quoniam BH = Γ + KM, quorum HΠ = KM, erit etiam ΥΧΦ = Γ.

Since BH is equal to Γ and KM together, and of these HΠ is equal to KM, the remaining gnomon ΥΧΦ is therefore equal to the remaining Γ.

Subtract equal regions from equal totals. Euclid then uses equal complements to turn this gnomon into the required parallelogram. The Latin compresses the relation into equations; the Greek keeps ‘remaining’ twice. The companion I.43/II.5 workshop operates that earlier argument.

λοιπὸς … λοιπῷ / erit etiam
remaining … to the remaining; the explicit remainder wording is compressed in the Latin equation

γνώμων / ΥΧΦ
gnomon, the L-shaped region; the Latin here uses only its three-letter name

VI.29 · opening problem

7. Change one word: exceed

Greek · p.166

Παρὰ τὴν δοθεῖσαν εὐθεῖαν τῷ δοθέντι εὐθυγράμμῳ ἴσον παραλληλόγραμμον παραβαλεῖν ὑπερβάλλον εἴδει παραλληλογράμμῳ ὁμοίῳ τῷ δοθέντι.

Heiberg’s Latin · p.167

Datae rectae datae figurae rectilineae aequale parallelogrammum adplicare excedens figura parallelogramma simili datae.

To apply to the given straight line a parallelogram equal to the given rectilinear figure, exceeding it by a parallelogram similar to the given one.

Compare ὑπερβάλλον / excedens with ἐλλεῖπον / deficiens in VI.28. The beyond-end region is now part of the applied area. For target 45 the falling-short family has no fit, but the exceeding family fits at t = 3: (12 + 3) × 3 = 45. Changing the mode changes the problem’s geometric condition.

ὑπερβάλλον / excedens
exceeding, extending beyond; a different participle changes the construction

ὁμοίῳ τῷ δοθέντι / simili datae
similar to the given one; the requirement on shape remains

VI.29 · construction on the half

8. Use the sum instead of the difference

Greek · p.168

Τετμήσθω ἡ AB δίχα κατὰ τὸ E, καὶ ἀναγεγράφθω ἀπὸ τῆς EB τῷ Δ ὅμοιον καὶ ὁμοίως κείμενον παραλληλόγραμμον τὸ BZ, καὶ συναμφοτέροις μὲν τοῖς BZ, Γ ἴσον, τῷ δὲ Δ ὅμοιον καὶ ὁμοίως κείμενον τὸ αὐτὸ συνεστάτω τὸ HΘ.

Heiberg’s Latin · p.169

secetur AB in duas partes aequales in puncto E, et in EB figurae Δ simile et similiter positum construatur parallelogrammum BZ, et BZ + Γ magnitudini aequale, parallelogrammo Δ autem simile et similiter positum idem construatur HΘ.

Bisect AB at E, and on EB construct BZ similar and similarly placed to Δ; construct HΘ equal to BZ and Γ together, and at the same time similar and similarly placed to Δ.

Both constructions begin with the half-line. VI.28 constructs an area difference; VI.29 constructs a sum. Our corresponding normalized identity is $(t + a/2)^2 = a^2/4 + c/k$. The right side is positive for every positive target; the positive solution has no counterpart to VI.28’s area ceiling.

δίχα / in duas partes aequales
in two equal parts; bisect the line

συναμφοτέροις / BZ + Γ magnitudini
to both together; the Latin writes the sum explicitly

VI.29 · the gnomon remainder

9. Subtract what is common

Greek · p.168

κοινὸν ἀφῃρήσθω τὸ ΕΛ· λοιπὸς ἄρα ὁ ΨΧΦ γνώμων τῷ Γ ἐστιν ἴσος.

Heiberg’s Latin · p.169

subtrahatur, quod commune est, ΕΛ. itaque est ΨΧΦ = Γ.

Subtract the common ΕΛ; the remaining gnomon ΨΧΦ is therefore equal to Γ.

ἀφῃρήσθω and subtrahatur direct the subtraction. The source then uses equal complements and adds a common region to obtain the applied parallelogram. An algebraic positive root models the resulting relation; finding it numerically does not perform every construction in Euclid’s proof.

κοινόν / quod commune est
what is common to the two totals

ἀφῃρήσθω / subtrahatur
let it be subtracted; a passive construction command

Keep and reopen the work

Save editable JSON · Save exact CSV · Save font-embedded SVG · Save source identities


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Reopen these settings in the app. This external link needs a connection. This reading’s text, diagrams, images, font, native downloads and licenses need none.

Fifteen whole source pages

Euclid, *Euclidis opera omnia*, vol. II, edited and translated into Latin by J. L. Heiberg. Leipzig: B. G. Teubner, 1884. Printed pp.158–171 include VI.27–29 and adjacent text. Title: PDF p.7. Numbered print pages plus 26 give the one-based PDF page. Complete original volume.

Line wrapping and line-end hyphens are joined, theta glyphs normalized, apparatus and superscript references omitted in selections. Editorial square brackets remain. Latin geometric letters use Greek Unicode; u/v spelling remains. The original images retain all marks and apparatus. Source PDF SHA-256: cd24f03c6e4e6d5f30a908d144c8f0f7003e99813334edb3a4107d748cb41043.

EUCLIDIS

ELEMENTA.

EDIDIT ET LATINE INTERPRETATUS EST

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LIPSIAE

IN AEDIBUS B. G. TEUBNERI.

MDCCCLXXXIV.

ΑΚ, οὕτως ἡ *ΗΑ* πρὸς τὴν *ΑΕ*. ἡ *ΗΑ* ἄρα πρὸς ἑκατέραν τῶν *ΑΚ*, *ΑΕ* τὸν αὐτὸν ἔχει λόγον. ἴση ἄρα ἐστὶν ἡ *ΑΕ* τῇ *ΑΚ* ἡ ἐλάττων τῇ μείζονι· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα οὖκ ἐστι περὶ τὴν αὐτὴν διάμετρον τὸ *ΑΒΓΔ* τῷ *ΑΖ*· περὶ τὴν αὐτὴν ἄρα ἐστὶ διάμετρον τὸ *ΑΒΓΔ* παραλληλόγραμμον τῷ *ΑΖ* παραλληλογράμμῳ.

Ἐὰν ἄρα ἀπὸ παραλληλογράμμου παραλληλόγραμμον ἀφαιρεθῇ ὅμοιόν τε τῷ ὅλῳ καὶ ὁμοίως κείμενον κοινὴν γωνίαν ἔχον αὐτῷ, περὶ τὴν αὐτὴν διάμετρόν ἐστι τῷ ὅλῳ· ὅπερ ἔδει δεῖξαι.

κξ′.

Πάντων τῶν παρὰ τὴν αὐτὴν εὐθεῖαν παραβαλλομένων παραλληλογράμμων καὶ ἐλλειπόντων εἴδεσι παραλληλογράμμοις ὁμοίοις τε καὶ ὁμοίως κειμένοις τῷ ἀπὸ τῆς ἡμισείας ἀναγραφομένῳ μέγιστόν ἐστι τὸ ἀπὶ τῆς ἡμισείας παραβαλλόμενον [παραλληλόγραμμον] ὅμοιον ὂν τῷ ἐλλείμματι.

Ἔστω εὐθεῖα ἡ *ΑΒ* καὶ τετμήσθω δίχα κατὰ τὸ *Γ*, καὶ παραβεβλήσθω παρὰ τὴν *ΑΒ* εὐθεῖαν τὸ *ΑΔ* παραλληλόγραμμον ἐλλεῖπον εἴδει παραλληλογράμμῳ τῷ *ΔΒ* ἀναγραφέντι ἀπὸ τῆς ἡμισείας τῆς *ΑΒ*, τουτέστι τῆς *ΓΒ*· λέγω, ὅτι πάντων τῶν παρὰ τὴν *ΑΒ* παραβαλλομένων παραλληλογράμμων καὶ ἐλλειπόντων εἴδεσι [παραλληλογράμμοις] ὁμοίοις τε καὶ ὁμοίως κειμένοις

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1. *ΑΚ*] P; *ΑΕΚ*, E in ras., F; *ΑΕ* V. *ΑΕ*] *ΑΒ* P, corr. m. rec.; *ΑΚ* V. ἄρα] om. P. 3. *ΑΕ*] *ΑΚ* PFBp, V m. 2. *ΑΚ*] *ΑΕ* PBFp, V m. 2. ἐλάτων F, corr. m. 2. 4. οὖκ] (alt.) om. BVp. ἐστιν PFB. 5. *ΑΖ*] Pφ; *ΑΘ*

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$HA : AK = HA : AE$. ergo HA ad utramque AK, AE eandem rationem habet. quare $AE = AK$ [V, 9] minor maiori; quod fieri non potest. quare fieri non potest, ut $AB\Gamma\Delta$, AZ circum eandem diametrum posita non sint. ergo parallelogramma $AB\Gamma\Delta$, AZ circum eandem diametrum posita sunt.

Ergo si a parallelogrammo aufertur parallelogrammum toti simile et similiter positum et communem angulum habens, circum eandem diametrum positum est ac totum; quod erat demonstrandum.

XXVII.

Omnium parallelogrammorum eidem rectae adplicatorum et deficientium figuris parallelogrammis similibus et similiter positis ei, quae in dimidia describitur, maximum est parallelogrammum dimidiae adplicatum defectui simile.

Sit recta AB et in duas partes aequales secetur in Γ, et rectae AB adplicetur parallelogrammum $A\Delta$ deficiens figura parallelogramma ΔB in dimidia rectae AB, hoc est in ΓB, descripta. dico, omnium parallelogrammorum rectae AB adplicatorum et figuris

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BVp. 6. *ἐστίν* P. 10. *ἔχον γωνίαν* V. *αὐτήν*] supra m. 1 p. 12. *λ΄* Fp. 17. *τε ἐστι* p. 18. *παραλαμβανόμενον* P; corr. m. rec. *παραλληλόγραμμον*] m. rec. P. *ὅμοιον*] corr. ex *ὅμοι* P. 19. *ὃν τῷ*] *ὃν τό φ* in ras. *ἐλλείματι* p. 21. *τήν*] *τὴν αὐτήν* P. *AΔ*] *Δ* in ras. m. 2 V; *AB φ*. 23. *ΔB*] *ΔΘ φ* (non F). Post hoc uocab. add. Theon: *ὁμοίῳ τε καὶ ὁμοίως ἀναγραφέντι* (F; pro *ὁμοίῳ* Bp*φ*, V m. 2 hab. *ὅμοιον*; pro *ἀναγραφέντι* Bp: *ἀναγραφέν*, V *κειμεν* seq. ras.; *-τι* in F punctis del.). *ἀναγραφέντι*] P; *τῷ* Theon (BFVp). *ἡμισείας*] *ἡμισείας ἀναγραφέντι* FV. *AB*] *AΔ φ* (non F). *τουτέστιν* P. 25. *εἴδεσι*] *φ* (aliud uerbum habuit F); *εἴδεσιν* P. 26. *παραλληλογράμμοις*] om. P.

Printed p.159 · PDF p.185 · SHA-256 ffd8077b6a20716f246ef1dfeb845cfc382e148158bb57a75bc467b98b467050

τῷ ΔΒ μέγιστόν ἐστι τὸ ΑΔ. παραβεβλήσθω γὰρ παρὰ τὴν ΑΒ εὐθεῖαν τὸ ΑΖ παραλληλόγραμμον ἐλλεῖπον εἴδει παραλληλογράμμῳ τῷ ΖΒ ὁμοίῳ τε καὶ ὁμοίως κειμένῳ τῷ ΔΒ· λέγω, ὅτι μεῖζόν ἐστι τὸ ΑΔ τοῦ ΑΖ.

Ἐπεὶ γὰρ ὅμοιόν ἐστι τὸ ΔΒ παραλληλόγραμμον τῷ ΖΒ παραλληλογράμμῳ, περὶ τὴν αὐτήν εἰσι διάμετρον. ἤχθω αὐτῶν διάμετρος ἡ ΔΒ, καὶ καταγεγράφθω τὸ σχῆμα.

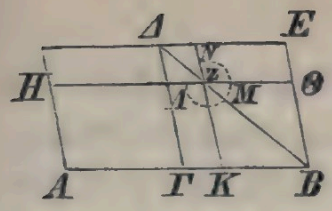
Ἐπεὶ οὖν ἴσον ἐστὶ τὸ ΓΖ τῷ ΖΕ, κοινὸν δὲ τὸ ΖΒ, ὅλον ἄρα τὸ ΓΘ ὅλῳ τῷ ΚΕ ἐστιν ἴσον. ἀλλὰ τὸ ΓΘ τῷ ΓΗ ἐστιν ἴσον, ἐπεὶ καὶ ἡ ΑΓ τῇ ΓΒ. καὶ τὸ ΗΓ ἄρα τῷ ΕΚ ἐστιν ἴσον. κοινὸν προσκείσθω τὸ ΓΖ· ὅλον ἄρα τὸ ΑΖ τῷ ΛΜΝ γνώμονί ἐστιν ἴσον· ὥστε τὸ ΔΒ παραλληλόγραμμον, τουτέστι τὸ ΑΔ, τοῦ ΑΖ παραλληλογράμμου μεῖζόν ἐστιν.

Πάντων ἄρα τῶν παρὰ τὴν αὐτὴν εὐθεῖαν παραβαλλομένων παραλληλογράμμων καὶ ἐλλειπόντων εἴδεσι παραλληλογράμμοις ὁμοίοις τε καὶ ὁμοίως κειμένοις τῷ ἀπὸ τῆς ἡμισείας ἀναγραφομένῳ μέγιστόν ἐστι τὸ ἀπὶ τῆς ἡμισείας παραβληθέν· ὅπερ ἔδει δεῖξαι.

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1. τῷ] τό F. παραβελήσθω p. 2. ΑΒ] Β e corr. m. 1 p. 3. παραλληλογράμω p. 7. περὶ ἄρα τήν Bp. 10. ἴσον] supra m. 1 V. ΖΕ] corr. ex ΖΘ m. rec. P. δέ] P; προσκείσθω Theon (BFVp). 11. ΓΘ] e corr. P m. rec. ΚΕ] corr. ex ΚΘ m. rec. P. 12. ΓΘ] corr. ex ΓΕ P m. rec. 13. ΓΒ] PF; ἐστιν ἴση supra add. V; ΓΒ ἴση ἐστίν Bp. ΕΚ] e corr. P m. rec. 14. ὅλον] seq. ras. 2—3 litt. F. 16. ΑΖ] inter Α et Ζ ras. 1 litt. F. 17. ἐστι B. 18. αὐτήν] om. p. 19. παραλληλογράμμων — 22: δεῖξαι] καὶ τὰ ἑξῆς p. 22. δεῖξαι] seq. in omnibus codd. demonstratio alia, quam in appendicem reiecimus; u. p. 161 not. 2.

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similibus et similiter positis figurae ΔB deficientium maximum esse $A\Delta$. adplicetur enim rectae AB parallelogrammum AZ deficiens figura parallelogramma ZB simili et similiter posita figurae ΔB. dico, esse $A\Delta > AZ$.

nam quoniam $\Delta B \sim ZB$, circum eandem diametrum sunt posita [prop. XXVI]. ducatur eorum diametrus ΔB, et describatur figura.[1]) iam quoniam $\Gamma Z = ZE$ [I, 43] et commune est ZB, erit $\Gamma\Theta = KE$. sed $\Gamma\Theta = \Gamma H$, quoniam $A\Gamma = \Gamma B$ [prop. I]. quare etiam $H\Gamma = EK$. commune adiiciatur ΓZ. itaque $AZ = \Delta MN$. quare $\Delta B > AZ$, h. e. $A\Delta > AZ$.

Ergo omnium parallelogrammorum eidem rectae adplicatorum et deficientium figuris parallelogrammis similibus et similiter positis ei, quae in dimidia describitur, maximum est, quod dimidiae adplicatur; quod erat demonstrandum.[2])

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1) H. e. producantur HZ ad Θ et KZ usque ad ΔE; cfr. II, 7, 8.

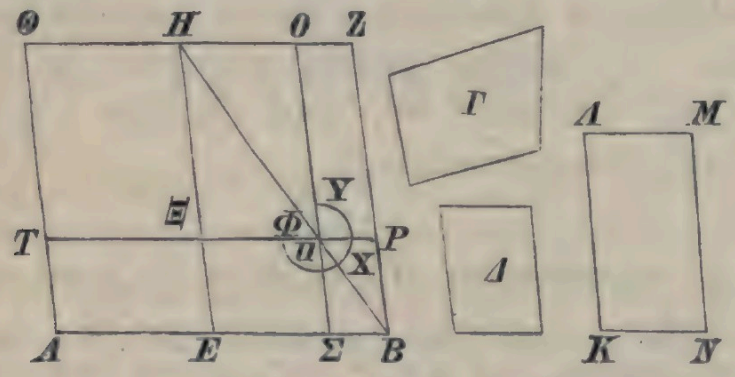
2) Itaque is solus casus tractatur, ubi $AK > A\Gamma$, nec opus est alterum, ubi $AK < A\Gamma$, propria demonstratione ostendere nec hoc moris est Euclidis. sane in codd. omnibus additur demonstratio huius quoque casus. sed apertissime interpolata est; nam primum ante lin. 18 sq., non post eas inserenda erat, deinde iam ab initio in praeparatione duo casus respiciendi erant nec hoc unquam neglexit Euclides, ubi plures casus habet; ita etiam in altero casu eaedem litterae, quae in priore, usurpatae essent, quod iure postulat Simsonus p. 380. Campanus VI, 26 utrumque casum demonstrat.

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κη'.

Παρὰ τὴν δοθεῖσαν εὐθεῖαν τῷ δοθέντι εὐθυγράμμῳ ἴσον παραλληλόγραμμον παραβαλεῖν ἐλλεῖπον εἴδει παραλληλογράμμῳ ὁμοίῳ τῷ δοθέντι· δεῖ δὲ τὸ διδόμενον εὐθύγραμμον [ᾧ δεῖ ἴσον παραβαλεῖν] μὴ μεῖζον εἶναι τοῦ ἀπὸ τῆς ἡμισείας ἀναγραφομένου ὁμοίου τῷ ἐλλείμματι [τοῦ τε ἀπὸ τῆς ἡμισείας καὶ ᾧ δεῖ ὅμοιον ἐλλείπειν].

Ἔστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ AB, τὸ δὲ δοθὲν εὐθύγραμμον, ᾧ δεῖ ἴσον παρὰ τὴν AB παραβαλεῖν, τὸ Γ μὴ μεῖζον [ὂν] τοῦ ἀπὸ τῆς ἡμισείας τῆς AB ἀναγραφομένου ὁμοίου τῷ ἐλλείμματι, ᾧ δὲ δεῖ ὅμοιον



ἐλλείπειν, τὸ Δ· δεῖ δὴ παρὰ τὴν δοθεῖσαν εὐθεῖαν τὴν AB τῷ δοθέντι εὐθυγράμμῳ τῷ Γ ἴσον παραλληλόγραμμον παραβαλεῖν ἐλλεῖπον εἴδει παραλληλογράμμῳ ὁμοίῳ ὄντι τῷ Δ.

Τετμήσθω ἡ AB δίχα κατὰ τὸ E σημεῖον, καὶ ἀναγεγράφθω ἀπὸ τῆς EB τῷ Δ ὅμοιον καὶ ὁμοίως

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1. κη'] om. F; λβ' p. 2. εὐθεῖαν] mg. m. 1 P. 3. παραβάλλειν V. 5. τῷ] ὄντι τῷ V. δέ] δή PBFVp;

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XXVIII.

Datae rectae datae figurae rectilineae aequale parallelogrammum adplicare deficiens figura parallelogramma datae simili. oportet autem, figuram rectilineam datam[1]) maiorem non esse figura in dimidia recta descripta defectui simili.[2])

Sit data recta AB, et data figura rectilinea, cui aequalem figuram rectae AB adplicare oportet, Γ non maior figura in dimidia AB descripta simili defectui, ea autem, cui similem figuram deficere oportet, sit Δ. oportet igitur datae rectae AB datae figurae rectilineae Γ aequale parallelogrammum adplicare deficiens figura parallelogramma simili figurae Δ.

secetur enim AB in duas partes aequales in puncto E, et in EB describatur figurae Δ similis et

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1) Uerba a Theone lin. 6 interpolata ideo parum necessaria sunt, quod *τὸ διδόμενον εὐθύγραμμον* ad *τῷ δοθέντι* (sc. *εἴδει*) lin. 5 referri non possunt, sed necessario a quouis lectore ad *τῷ δοθέντι εὐθυγράμμῳ* lin. 2 trahuntur.

2) Hunc *διορισμόν* statim praebet prop. 27. — Campanum VI, 27: „quod secundum eiusdem suum esse parallelogrammo super dimidiam datae lineae collocato minime maius existat“ non intellego, uidetur tamen potius cum P consentire.

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corr. Augustus. 6. *ᾧ δεῖ ἴσον παραβαλεῖν*] add. Theon (BFVp); m. rec. P. *παραβάλλειν* FV. 7. *ἀναγραφομένου*] P; *παραβαλλομένου* Theon (BFVp). *ὁμοίου*] P; *ὁμοίων ὄντων* Theon (BFVp), P m. rec. *τῷ ἐλλείμματι*] P; *τῶν ἐλλειμμάτων* Theon (BFVp), P m. rec. 8. *τοῦ τε* — 9: *ἐλλείπειν*] add. Theon (BFVp); m. rec. P. 12. *ὅν*] om. P. *τοῦ*] *τῷ φ*. *τῆς AB*] P; om. Theon (BFVp). 13. *ἀναγραφομένου*] P; *παραβαλλομένου* Theon (BFVp). *ὁμοίου τῷ ἐλλείμματι*] P; *ὁμοίων ὄντων τῶν ἐλλειμμάτων* Theon (BFVp). 18. *τό E*] euan. F.

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κείμενον τὸ *EBZH*, καὶ συμπεπληρώσθω τὸ *AH* παραλληλόγραμμον.

Εἰ μὲν οὖν ἴσον ἐστὶ τὸ *AH* τῷ Γ, γεγονὸς ἂν εἴη τὸ ἐπιταχθέν· παραβέβληται γὰρ παρὰ τὴν δοθεῖσαν εὐθεῖαν τὴν *AB* τῷ δοθέντι εὐθυγράμμῳ τῷ Γ ἴσον παραλληλόγραμμον τὸ *AH* ἐλλεῖπον εἴδει παραλληλογράμμῳ τῷ *HB* ὁμοίῳ ὄντι τῷ *Δ*. εἰ δὲ οὔ, μεῖζόν ἔστω τὸ *ΘE* τοῦ Γ. ἴσον δὲ τὸ *ΘE* τῷ *HB*· μεῖζον ἄρα καὶ τὸ *HB* τοῦ Γ. ᾧ δὴ μεῖζόν ἐστι τὸ *HB* τοῦ Γ, ταύτῃ τῇ ὑπεροχῇ ἴσον, τῷ δὲ *Δ* ὅμοιον καὶ ὁμοίως κείμενον τὸ αὐτὸ συνεστάτω τὸ *KΛMN*. ἀλλὰ τὸ *Δ* τῷ *HB* [ἐστιν] ὅμοιον· καὶ τὸ *KM* ἄρα τῷ *HB* ἐστιν ὅμοιον. ἔστω οὖν ὁμόλογος ἡ μὲν *KΛ* τῇ *HE*, ἡ δὲ *ΛM* τῇ *HZ*. καὶ ἐπεὶ ἴσον ἐστὶ τὸ *HB* τοῖς Γ, *KM*, μεῖζον ἄρα ἐστὶ τὸ *HB* τοῦ *KM*· μείζων ἄρα ἐστὶ καὶ ἡ μὲν *HE* τῆς *KΛ*, ἡ δὲ *HZ* τῆς *ΛM*. κείσθω τῇ μὲν *KΛ* ἴση ἡ *HΞ*, τῇ δὲ *ΛM* ἴση ἡ *HO*, καὶ συμπεπληρώσθω τὸ *ΞHOΠ* παραλληλόγραμμον· ἴσον ἄρα καὶ ὅμοιόν ἐστι [τὸ *HΠ*] τῷ *KM* [ἀλλὰ τὸ *KM* τῷ *HB* ὅμοιόν ἐστιν]. καὶ τὸ *HΠ* ἄρα τῷ *HB* ὅμοιόν ἐστιν· περὶ τὴν αὐτὴν ἄρα διάμετρόν ἐστι τὸ *HΠ* τῷ *HB*. ἔστω αὐτῶν διάμετρος ἡ *HΠB*, καὶ καταγεγράφθω τὸ σχῆμα.

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1. *EBZH*] *BEZH* F? 2. Post παραλληλόγραμμον add. Theon: τὸ δὴ *AH* ἤτοι ἴσον ἐστὶ τῷ Γ ἢ μεῖζον αὐτοῦ διὰ τὸν διορισμόν (BVp, F mg. m. 1; pro διορισμόν habent FV ὁρισμόν; in V corr. m. 2). 3. ἐστίν P; in F cum τὸ *AH* euan. 6. *AH*] euan. F. 8. δέ] δ' F. ἔστω] PF; ἔσται Bp; ἐστι V. δὲ τό] δὲ τοῦ B. 9. τῷ] τό B. *HB*] *H* supra m. 1 V. δή] δὲ uel δεῖ B; δεῖ p. 12. ἐστιν] om. P. 13. *KM*] inter *K* et *M* una litt. (ε?) euan. F. 14. *KΛ*] *ΛK* Bp. 15. *HB*] e corr. m. 1 p. 17. *KΛ*] *ΛK* Bp.

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similiter posita $EBZH$ [prop. XVIII], et expleatur parallelogrammum AH. iam si $AH = \Gamma$, effectum erit propositum. nam datae rectae AB datae figurae rectilineae Γ aequale parallelogrammum adplicatum est AH deficiens figura parallelogramma HB simili figurae Δ. sin minus, sit $\Theta E > \Gamma$.[1]) sed $\Theta E = HB$. itaque $HB > \Gamma$. iam excessui, quo maius est HB figura Γ, aequale et parallelogrammo Δ simile et similiter positum idem construatur $K\Lambda MN$ [prop. XXV]. sed $\Delta \sim HB$. quare etiam $KM \sim HB$ [prop. XXI]. iam correspondeant inter se $K\Lambda$, HE et ΛM, HZ. et quoniam $HB = \Gamma + KM$, erit $HB > KM$. quare etiam $HE > K\Lambda$, $HZ > \Lambda M$.[2]) ponatur $H\Xi = K\Lambda$ et $HO = \Lambda M$, et expleatur parallelogrammum $\Xi HO\Pi$. itaque aequale et simile[3]) est parallelogrammo KM. quare etiam $H\Pi \sim HB$ [prop. XXI, cfr. lin. 13]. itaque $H\Pi$, HB circum eandem diametrum posita sunt [prop. XXVI]. sit eorum diametrus $H\Pi B$, et describatur figura [p. 161 not. 1].

1) Ex hypothesi; quare debuit esse *ἔσται* lin. 8, sed *ἔστω* ferri posse negare non ausim.

2) Nam per prop. 20 erit $HB : KM = HE^2 : K\Lambda^2 = HZ^2 : \Lambda M^2$. iam cum $HB > KM$, erit $HE^2 > K\Lambda^2$, $HZ^2 > \Lambda M^2$, h. e. $HE > K\Lambda$, $HZ > \Lambda M$.

3) Quia $HB \sim KM$, erit $\angle OH\Xi = K\Lambda M$. itaque $H\Pi$, KM aequiangula sunt. quare et similia sunt (def. 1) et aequalia (prop. 14). cfr. p. 144, 11.

τῇ μὲν ΚΛ] Bp; *τῇ ΚΛ μέν* PF; *μὲν τῇ ΚΛ* V. 18. *HO*] corr. ex *HΘ* m. rec. P; *O* e corr. m. 2 V; *HΘ* F? 20. *τὸ ΗΠ*] om. P. *τῷ*] e corr. P. *ἀλλὰ τὸ ΚΜ τῷ ΗΒ ὅμοιόν ἐστιν*] *τὸ ΗΠ*. *ἀλλὰ τὸ ΚΜ τῷ ΗΒ ὅμοιόν ἐστι* supra m. rec. P. *ΚΜ*] *Κ* in ras. m. 2 V. 21. *ἐστι* BVp. *ἐστι* BFV, comp. p.

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ἐπεὶ οὖν ἴσον ἐστὶ τὸ *BH* τοῖς *Γ*, *KM*, ὧν τὸ *HΠ* τῷ *KM* ἐστιν ἴσον, λοιπὸς ἄρα ὁ *ΥΧΦ* γνώμων λοιπῷ τῷ *Γ* ἴσος ἐστίν. καὶ ἐπεὶ ἴσον ἐστὶ τὸ *OP* τῷ *ΞΣ*, κοινὸν προσκείσθω τὸ *ΠB*· ὅλον ἄρα τὸ *OB* ὅλῳ τῷ *ΞB* ἴσον ἐστίν. ἀλλὰ τὸ *ΞB* τῷ *TE* ἐστιν ἴσον, ἐπεὶ καὶ πλευρὰ ἡ *AE* πλευρᾷ τῇ *EB* ἐστιν ἴση· καὶ τὸ *TE* ἄρα τῷ *OB* ἐστιν ἴσον. κοινὸν προσκείσθω τὸ *ΞΣ*· ὅλον ἄρα τὸ *TΣ* ὅλῳ τῷ *ΦΧΥ* γνώμονί ἐστιν ἴσον. ἀλλ' ὁ *ΦΧΥ* γνώμων τῷ *Γ* ἐδείχθη ἴσος· καὶ τὸ *TΣ* ἄρα τῷ *Γ* ἐστιν ἴσον.

Παρὰ τὴν δοθεῖσαν ἄρα εὐθεῖαν τὴν *AB* τῷ δοθέντι εὐθυγράμμῳ τῷ *Γ* ἴσον παραλληλόγραμμον παραβέβληται τὸ *ΣT* ἐλλεῖπον εἴδει παραλληλογράμμῳ τῷ *ΠB* ὁμοίῳ ὄντι τῷ *Δ* [ἐπειδήπερ τὸ *ΠB* τῷ *HΠ* ὅμοιόν ἐστιν]· ὅπερ ἔδει ποιῆσαι.

κθ'.

Παρὰ τὴν δοθεῖσαν εὐθεῖαν τῷ δοθέντι εὐθυγράμμῳ ἴσον παραλληλόγραμμον παραβαλεῖν ὑπερβάλλον εἴδει παραλληλογράμμῳ ὁμοίῳ τῷ δοθέντι.

Ἔστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ *AB*, τὸ δὲ δοθὲν εὐθύγραμμον, ᾧ δεῖ ἴσον παρὰ τὴν *AB* παραβαλεῖν, τὸ *Γ*, ᾧ δὲ δεῖ ὅμοιον ὑπερβάλλειν, τὸ *Δ*· δεῖ δὴ παρὰ τὴν *AB* εὐθεῖαν τῷ *Γ* εὐθυγράμμῳ ἴσον παραλληλόγραμμον παραβαλεῖν ὑπερβάλλον εἴδει παραλληλογράμμῳ ὁμοίῳ τῷ *Δ*.

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1. *BH*] in ras. m. 2 V; *HB* p. 2. *ἴσον ἐστίν* p. *λοιπόν* P, corr. m. rec. *ΥΧΦ*] *ΥΦΧ* P V. 3. *ἐστιν ἴσος* F. *ἐστίν*] *ἐστί* V, comp. p. *ἐστί*] *ἐστίν* P. 5. *OB*] euan. F. 6. *ἴσον ἐστίν* B. Ante *ἐπεί* add. φ: *ἐπϊ*. 7. *OB*] *O* in

iam quoniam $BH = \Gamma + KM$, quorum $H\Pi = KM$, erit etiam $\Upsilon X\Phi = \Gamma$. et quoniam $OP = \Xi\Sigma$ [I, 43], commune adiiciatur ΠB. itaque $OB = \Xi B$. sed $\Xi B = TE$, quoniam $AE = EB$ [prop. I]. quare etiam $TE = OB$. commune adiiciatur $\Xi\Sigma$. itaque $T\Sigma = \Phi X\Upsilon$. sed demonstratum est, esse $\Phi X\Upsilon = \Gamma$. quare etiam $T\Sigma = \Gamma$.

Ergo datae rectae AB datae figurae rectilineae Γ aequale parallelogrammum adplicatum est ΣT deficiens figura parallelogramma ΠB, quae figurae Δ similis est[1]); quod oportebat fieri.

XXIX.

Datae rectae datae figurae rectilineae aequale parallelogrammum adplicare excedens figura parallelogramma simili datae.

Sit data recta AB, data autem figura rectilinea, cui aequalem figuram rectae AB adplicare oportet, sit Γ, ea autem, cui similem figuram excedere oportet, sit Δ. oportet igitur rectae AB figurae rectilineae Γ aequale parallelogrammum adplicare excedens figura parallelogramma simili figurae Δ.

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1) Nam $\Pi B \sim HB$ (prop. 24) $\sim \Delta$. uerba *ἐπειδήπερ — ἐστιν*, ubi sine causa de $H\Pi$ mentio iniicitur, spuria sunt. alia res est p. 170, 7.

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ras. m. 2 V. 8. $T\Sigma$] TB corr. ex $T\Gamma$ m. 1 p. 9. *ἀλλά* Bp. 10. $T\Sigma$] $A\Pi$ P. 11. *ἄρα*] om. F. 13. Supra ΣT ras. est in V. 14. *τῷ*] (tert.) postea insert. m. 1 F. 16. *κθ΄*] *λγ΄* p et F, corr. m. rec. 18. *παραλληλόγραμμον*] *παραλληλο-* sustulit resarcinatio in F. 22. *δεῖ*] *δή* F p. 23. *ὑπερβαλεῖν* F. *δεῖ δή*] sustulit lac. pergameni F. 24. *παρά — εὐθυγράμμῳ*] mg. m. 1 F. *ἴσον*] in ras. F.

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Τετμήσθω ἡ AB δίχα κατὰ τὸ E, καὶ ἀναγεγράφθω ἀπὸ τῆς EB τῷ $Δ$ ὅμοιον καὶ ὁμοίως κείμενον παραλληλόγραμμον τὸ BZ, καὶ συναμφοτέροις μὲν τοῖς BZ, $Γ$ ἴσον, τῷ δὲ $Δ$ ὅμοιον καὶ ὁμοίως κείμενον τὸ αὐτὸ συνεστάτω τὸ $HΘ$. ὁμόλογος δὲ ἔστω ἡ μὲν $KΘ$ τῇ $ZΛ$, ἡ δὲ KH τῇ ZE. καὶ ἐπεὶ μεῖζόν ἐστι τὸ $HΘ$ τοῦ ZB, μεῖζων ἄρα ἐστὶ καὶ ἡ μὲν $KΘ$ τῆς $ZΛ$, ἡ δὲ KH τῆς ZE. ἐκβεβλήσθωσαν αἱ $ZΛ$, ZE, καὶ τῇ μὲν $KΘ$ ἴση ἔστω ἡ $ZΛM$, τῇ δὲ KH ἴση ἡ ZEN, καὶ συμπεπληρώσθω τὸ MN· τὸ MN ἄρα τῷ $HΘ$ ἴσον τέ ἐστι καὶ ὅμοιον. ἀλλὰ τὸ $HΘ$ τῷ $EΛ$ ἐστιν ὅμοιον· καὶ τὸ MN ἄρα τῷ $EΛ$ ὅμοιόν ἐστιν· περὶ τὴν αὐτὴν ἄρα διάμετρόν ἐστι τὸ $EΛ$ τῷ MN. ἤχθω αὐτῶν διάμετρος ἡ $ZΞ$, καὶ καταγεγράφθω τὸ σχῆμα.

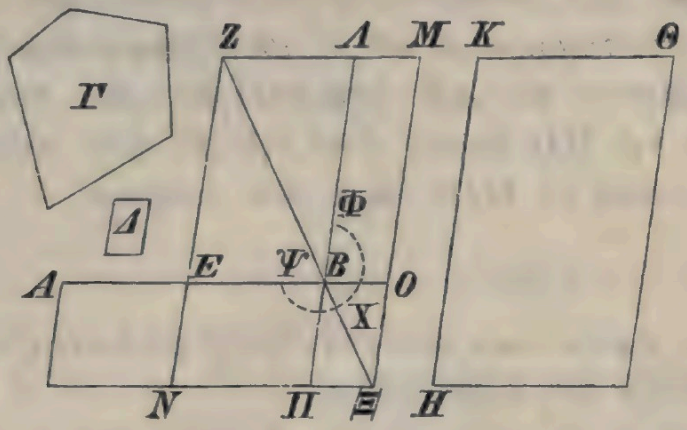
Ἐπεὶ ἴσον ἐστὶ τὸ $HΘ$ τοῖς $EΛ$, $Γ$, ἀλλὰ τὸ $HΘ$ τῷ MN ἴσον ἐστίν, καὶ τὸ MN ἄρα τοῖς $EΛ$, $Γ$ ἴσον ἐστίν. κοινὸν ἀφῃρήσθω τὸ $EΛ$· λοιπὸς ἄρα ὁ $ΨXΦ$ γνώμων τῷ $Γ$ ἐστιν ἴσος. καὶ ἐπεὶ ἴση ἐστὶν ἡ AE τῇ EB, ἴσον ἐστὶ καὶ τὸ AN τῷ NB, τουτέστι τῷ $ΛO$. κοινὸν προσκείσθω τὸ $EΞ$· ὅλον ἄρα τὸ

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3. BZ] corr. ex HZ m. 2 V. 4. BZ, $Γ$] Z et $Γ$ e corr. p; HZ, $Γ$ V. $Δ$] e corr. F. 5. $HΘ$] PF; $HΘ$· ὅμοιον ἄρα ἐστὶ τὸ $HΘ$ τῷ ZB Bp, V mg. m. 2. 6. ZE] EZ F. 8. $KΘ$] $ΘK$ F. 10. KH] corr. ex KB m. rec. P. 11. τε] om. V. ἐστιν P. 12. τό] (alt.) τῷ F, sed corr. 13. $EΛ$] $Λ$ F. ἐστιν ὅμοιον V. ἐστιν] P, comp. p; ἐστι BFV. 14. ἐστι] supra F. αὐτῶν] αὐτῶν ἡ V. 16. ἐπεὶ οὖν FV. τό] (prius) τῷ F. 17. ἐστί P, BV, comp. p. 18. ἐστί BV, comp. p. $EΛ$] mutat. in $ΘΛ$ m. 1 F. 20. AE] in ras. m. 2 V. τουτέστιν P; comp. p. 21. $ΛO$] O e corr. m. 1 F.

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secetur AB in duas partes aequales in puncto E, et in EB figurae Δ simile et similiter positum construatur parallelogrammum BZ, et $BZ + \Gamma$ magni-



tudini aequale, parallelogrammo Δ autem simile et similiter positum idem construatur $H\Theta$ [prop. XXV]. correspondeant[1]) autem $K\Theta$, $Z\Lambda$ et KH, ZE. et quoniam $H\Theta > ZB$, erit etiam $K\Theta > Z\Lambda$ et $KH > ZE$ [p. 165 not. 2]. producantur $Z\Lambda$, ZE, et sit

$$Z\Lambda M = K\Theta,\ ZEN = KH,$$

et expleatur parallelogrammum MN. itaque MN et aequale et simile est parallelogrammo $H\Theta$ [p. 165 not. 3]. sed $H\Theta \sim E\Lambda$. quare etiam $MN \sim E\Lambda$ [prop. XXI]. itaque circum eandem diametrum posita sunt $E\Lambda$, MN [prop. XXVI]. ducatur eorum diametrus $Z\Xi$, et describatur figura.

iam quoniam $H\Theta = E\Lambda + \Gamma$ et $H\Theta = MN$, erit etiam $MN = E\Lambda + \Gamma$. subtrahatur, quod commune est, $E\Lambda$. itaque est $\Psi X\Phi = \Gamma$. et quoniam $AE = EB$, erit $AN = NB = \Lambda O$ [I, 43]. commune adiiciatur

1) Sc. in ΘH, $E\Lambda$ parallelogrammis, quae figurae Δ similia sunt; unde etiam inter se similia sunt (prop. 21).

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ΑΞ ἴσον ἐστὶ τῷ ΦΧΨ γνώμονι. ἀλλὰ ὁ ΦΧΨ γνώμων τῷ Γ ἴσος ἐστίν· καὶ τὸ ΑΞ ἄρα τῷ Γ ἴσον ἐστίν.

Παρὰ τὴν δοθεῖσαν ἄρα εὐθεῖαν τὴν ΑΒ τῷ δοθέντι εὐθυγράμμῳ τῷ Γ ἴσον παραλληλόγραμμον παραβέβληται τὸ ΑΞ ὑπερβάλλον εἴδει παραλληλογράμμῳ τῷ ΠΟ ὁμοίῳ ὄντι τῷ Δ, ἐπεὶ καὶ τῷ ΕΔ ἐστιν ὅμοιον τὸ ΟΠ· ὅπερ ἔδει ποιῆσαι.

λ΄.

Τὴν δοθεῖσαν εὐθεῖαν πεπερασμένην ἄκρον καὶ μέσον λόγον τεμεῖν.

Ἔστω ἡ δοθεῖσα εὐθεῖα πεπερασμένη ἡ ΑΒ· δεῖ δὴ τὴν ΑΒ εὐθεῖαν ἄκρον καὶ μέσον λόγον τεμεῖν.

Ἀναγεγράφθω ἀπὸ τῆς ΑΒ τετράγωνον τὸ ΒΓ, καὶ παραβεβλήσθω παρὰ την ΑΓ τῷ ΒΓ ἴσον παραλληλόγραμμον τὸ ΓΔ ὑπερβάλλον εἴδει τῷ ΑΔ ὁμοίῳ τῷ ΒΓ.

Τετράγωνον δέ ἐστι τὸ ΒΓ· τετράγωνον ἄρα ἐστὶ καὶ τὸ ΑΔ. καὶ ἐπεὶ ἴσον ἐστὶ τὶ ΒΓ τῷ ΓΔ, κοινὸν ἀφῃρήσθω τὸ ΓΕ· λοιπὸν ἄρα τὸ ΒΖ λοιπῷ τῷ ΑΔ ἐστιν ἴσον. ἔστι δὲ αὐτῷ καὶ ἰσογώνιον· τῶν ΒΖ, ΑΔ ἄρα ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰς ἴσας γωνίας· ἔστιν ἄρα ὡς ἡ ΖΕ πρὸς τὴν ΕΔ, οὕτως ἡ ΑΕ πρὸς τὴν ΕΒ. ἴση δὲ ἡ μὲν ΖΕ τῇ ΑΒ, ἡ δὲ ΕΔ τῇ ΑΕ. ἔστιν ἄρα ὡς ἡ ΒΑ πρὸς

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1. ἀλλ᾽ F. 2. ἴσος] ἴσον φ (non F). ἐστίν] F, comp. p; ἐστί PBV. 3. ἐστί B. 4. ἄρα] supra comp. F. εὐθεῖάν ἐστι F. 7. τῷ] (alt.) τό F, et V, corr. m. 2. 9. λδ΄] p; F, sed corr. m. rec. 11. τεμεῖν] supra scr. ν m. 1 F. 14. γὰρ ἀπό F V. Post *AB* ras. magna F. 15. *AΓ*] corr. ex *AB* m. 1 F. 20. BZ] corr. ex *BΓ* m. 1 p. 21. τῷ] τό φ

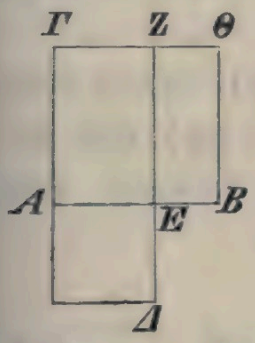
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$E\Xi$. itaque $A\Xi = \Phi X \Psi$. sed $\Phi X \Psi = \Gamma$. quare etiam $A\Xi = \Gamma$.

Ergo datae rectae AB datae figurae rectilineae Γ aequale adplicatum est parallelogrammum $A\Xi$ excedens figura parallelogramma ΠO, quae similis est figurae Δ, quia $O\Pi \sim E\Delta$ [prop. XXIV]; quod oportebat fieri.

XXX.

Datam rectam terminatam secundum rationem extremam ac mediam secare.



Sit data recta terminata AB. oportet igitur rectam AB secundum extremam ac mediam rationem secare.

describatur enim in AB quadratum $B\Gamma$, et rectae $A\Gamma$ adplicetur parallelogrammum $\Gamma\Delta$ quadrato $B\Gamma$ aequale et excedens figura $A\Delta$ simili figurae $B\Gamma$ [prop. XXIX]. quadratum autem est $B\Gamma$; itaque etiam $A\Delta$ quadratum est. et quoniam $B\Gamma = \Gamma\Delta$, subtrahatur, quod commune est, ΓE. quare $BZ = A\Delta$. uerum etiam aequiangulum ei est.[1]) quare in parallelogrammis BZ, $A\Delta$ latera aequales angulos comprehendentia in contraria proportione sunt [prop. XIV]. itaque $ZE : E\Delta = AE : EB$. sed $ZE = AB$[2]) et

1) Nam utrumque rectangulum est.
2) Nam $ZE = A\Gamma$ (I, 34) et $A\Gamma = AB$.

(non F). ἴσον ἐστίν F. 23. τήν] om. BFp. 24. AE] AB φ. τήν] om. BFp. ZE τῇ $A\Gamma$, τουτέστι τῇ AB Theon (BFVp). 25. AE] AB φ.

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The experiment operates a coordinate family of parallelograms; it does not perform every compass-and-straightedge step or reconstruct ancient instruments. Numerical graph pixels and approximate root labels are display aids. Fractions determine exact areas, root type and certified bounds. Input limits: integer line 4–24, positive target at most 4096, denominator at most 1,000,000, shortfall 0–a or extension 0–128. Zero-area endpoints are labeled degeneracies.

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